Environmental Science and Pollution Research (2023) 30:85087–85101
https://doi.org/10.1007/s11356-023-28407-x

RESEARCH ARTICLE

Optimization of China's freight transportation structure based on adaptive genetic algorithm under the background of carbon peak

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Received: 1 September 2022 / Accepted: 19 June 2023 / Published online: 28 June 2023
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Abstract

Transportation is the key carbon emission source after energy supply and industrial production. Under the vision of carbon peak and carbon neutralization, the pressure of reducing carbon emissions in transportation will be greater in the future. This paper constructs a model that takes transportation carbon emission as the main target and freight transportation utility value as the auxiliary target. The constructed model satisfies the constraints of freight turnover in the whole society, freight economic and social benefits, and the ecological constraints of the freight system. With MATLAB, the freight turnover of roadways, railways, and waterways (excluding ocean transportation) in 2030 is solved by using the adaptive genetic algorithm. The results indicate that (I) compared with the current freight structure of China, the roadway freight sharing rate in 2030 will decrease by 8.07%, and the railway freight sharing rate and the waterway freight sharing rate (excluding ocean transportation) will increase by 0.93% and 7.13%, respectively. (II) After optimization, the energy consumption and carbon emission are reduced by 42,471,500 tons (10.3%) and 91,379,400 tons (10.2%) of standard coal, respectively. (III) The adaptive genetic algorithm outperforms the traditional genetic algorithm in terms of convergence speed and accuracy. (IV) As the weight coefficient of carbon emission increases, the utility value of freight transportation consistently decreases, and the sensitivity increases. Meanwhile, as the carbon emission weight coefficient increases, carbon emission keeps decreasing, and the sensitivity decreases.

Keywords AGA · Freight transportation · Carbon emission · Energy consumption. Carbon peak · Ecological

Introduction

In recent years, the issue of carbon emissions has attracted more and more attention from people all over the world (Solaymani 2019). On September 22, 2020, President Xi Jinping announced that China will increase its contribution to global autonomy and adopt more forceful policy measures during the general debate of the 75th session of the United Nations General Assembly. China aims to peak its carbon dioxide emissions by 2030 and become carbon neutral by 2060 (Li et al. 2022). It can be seen that China's next carbon emission reduction task is significant.

Currently, there is a huge potential for carbon reduction in the Chinese mode of freight transport (Gracht and Darkow 2016). It can be seen from Fig. 1 that the proportion of roadway freight ranks first all year long and far exceeds that of other modes of freight transportation. In 2020 and 2021, the proportions of roadway freight transportation exceeded 50%, and there was a growing trend. Meanwhile, roadway transport is known for its characteristic of high carbon intensity. The carbon emission per unit of roadway transport turnover is 19.65 times that of railway transport and 10.72 times that of waterway transport. This demonstrates that the proportion of roadway freight turnover is relatively high, and the carbon emission intensity of roadway freight unit turnover is high. Therefore, adjusting the roadway freight sharing rate reasonably can effectively reduce traffic carbon emissions (Lin and Wang 2021).

Concerning the adjustment of freight structure, our strategy is to plan economic and social benefits and rationally allocate and integrate transportation resources (Lu et al. 2023, Zuo et al. 2023). At present, China has issued

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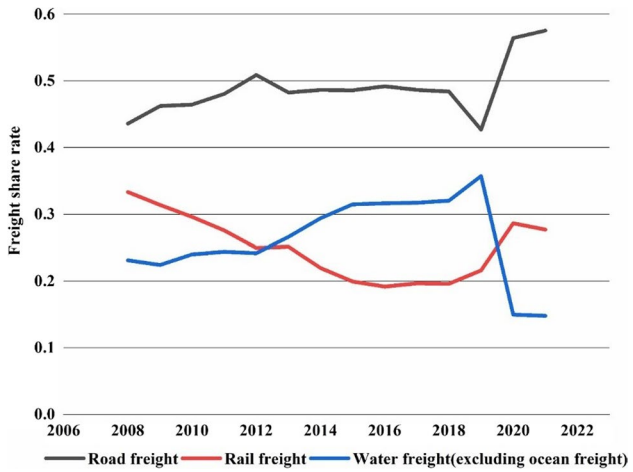


Fig. 1 The freight structure changes of roadway, railway, and waterway from 2008 to 2021 (excluding sea transportation)

relevant policies to limit the development of roadway freight transport. Though this can have a positive effect on carbon reduction, each mode of freight transport has different characteristics. In this case, it is impossible to blindly choose a mode of transport with low carbon emission intensity while ignoring other features of freight transport, such as safety, transport price, speed, and flexibility. Also, it is necessary to meet the needs of transportation turnover of the whole society, pay attention to the economic and social benefits of transportation, and consider the ecology of the transportation system. China's freight structure optimization problem needs to be considered from multiple objectives under multiple constraints.

For the multi-objective optimization problem of freight transportation structure adjustment, it is challenging to find a feasible global optimal solution. At present, the mainstream algorithms for multi-objective optimization problems include the traditional genetic algorithm (GA) (Vitorino et al. 2015), MOEA/D (Chen et al. 2022), intensity Pareto evolutionary algorithm (Suri and Vijay 2020), and adaptive genetic algorithm (AGA) (Xiong et al. 2023).

AGA is an improvement of the traditional GA, and by adaptively adjusting genetic parameters, it greatly improves the convergence accuracy and speed of the traditional GA. Compared with other intelligent algorithms, AGA has the

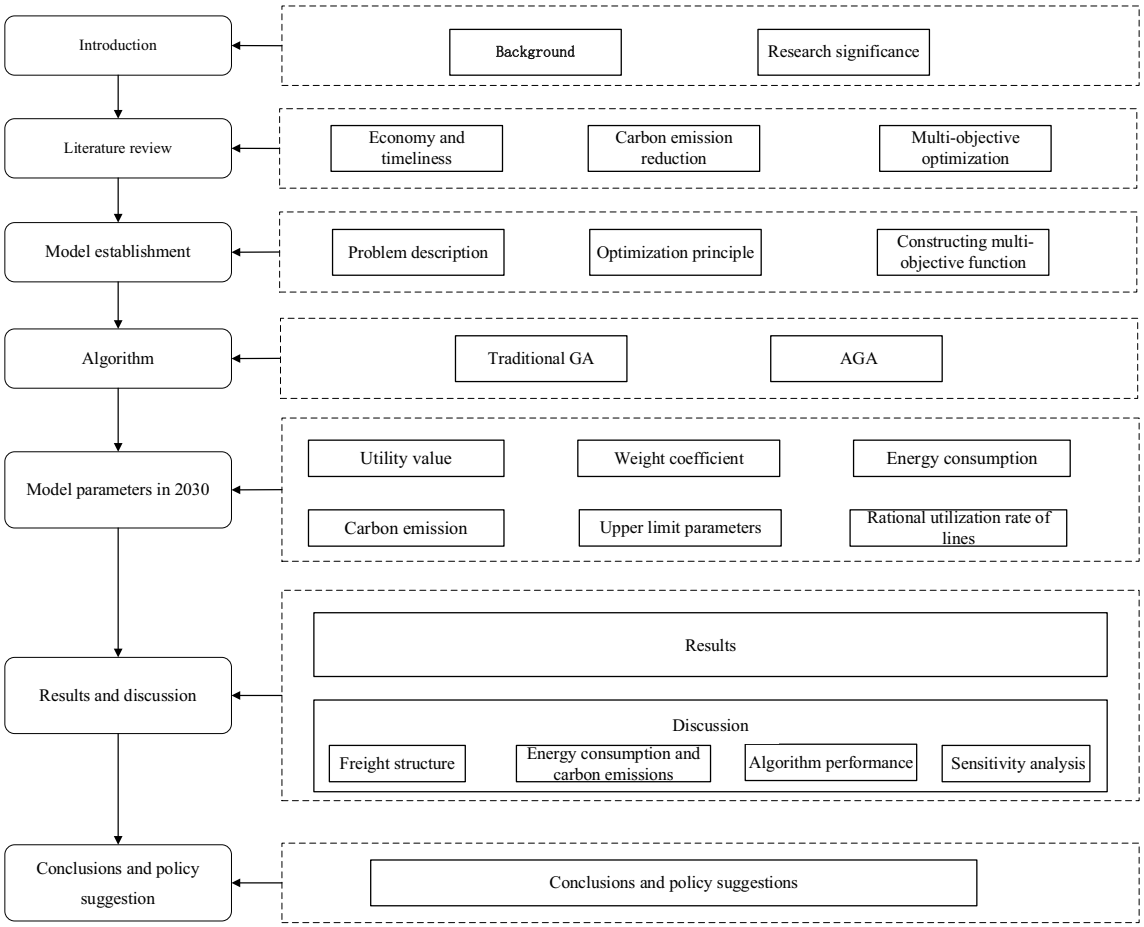


Fig. 2 Research roadmap

following advantages. Firstly, AGA directly takes the value of the objective function as the search message, which makes AGA a high degree of superiority. Secondly, AGA is based on probabilistic rules rather than definite rules, which makes the search process more flexible. Thirdly, AGA is easy to mix with other technologies. Compared with the traditional GA, AGA ensures the convergence of the genetic algorithm while maintaining population diversity. The genetic parameters of AGA are adaptive, which improves the convergence speed and accuracy of the traditional GA. Meanwhile, AGA has a wide range of applications, such as production scheduling (Low et al. 2013), automatic control (Ortiz-Quisbert et al. 2016), image processing (Buzylev and Shcherbakova 2021), and machine learning (Zhang et al. 2020).

The rest of this paper is organized as follows: "Literature review" section consists of three parts, including economy and timeliness, carbon emission reduction, and multi-objective optimization. "Model establishment" section mainly introduces problems, optimization principles, multi-objectives, and multi-constraints. "Algorithm" section presents the model-based algorithm design. In the "Determination of model parameters in 2030" section, the planning year is set to be 2030. "Results and discussions" section solves and analyzes the research results based on models and algorithms. The result analysis includes a comparative analysis of freight structure, energy consumption, and carbon emission, and algorithm performance and sensitivity analysis. Finally, "Conclusions and policy suggestions" section presents the conclusions, policy suggestions, and future directions (Fig. 2).

Literature review

Since there are few studies on the multi-objective optimization problem of freight transport structure, this paper summarizes relevant research on transport structure optimization from the perspective of generalized transport structure, which can be mainly divided into the following three categories.

The first category is the optimization of the freight structure, focusing on exploring transportation economy and timeliness. For example, taking transportation cost, transportation time, and goods attributes as the main factors, Holguin-Veras et al. (2021) used the discrete choice model to study the best choice of goods transportation both qualitatively and quantitatively according to the potential obstacles. Lei et al. (2014) established a multimodal transport scheme based on a mathematical model of transport cost, transport time, and transport risk, and then this scheme was solved with the particle swarm optimization algorithm and ant colony algorithm (PSACO). Gao et al. (2020) analyzed the coordination of freight modes from three aspects: coordination within a single freight mode system, coordination between different freight modes, and coordination between different regional freight modes and economies. The analysis results indicate that the coordination of a single freight transportation mode in four regions of our country is above 0.80, the coordination between waterway and air cargo is greater than 0.83, the coordination of various transport modes in the eastern region is higher than 0.78, and the overall coordination is better. Therefore, some scholars focus on exploring the economy and timeliness of the transportation structure.

The second category is to use intelligent simulation methods to investigate the transportation structure optimization problem with the goal of carbon emission reduction. Some studies construct models for solving the optimization problem. Li et al. (2019a) constructed a multi-objective optimization model of urban comprehensive traffic structure under resource constraints from the perspective of passenger transport, and they solved and compared the model by ideal point method, linear weighting method, and hierarchical sequence method. Qiang et al. (2018) established an energy consumption model to optimize the urban traffic structure of Harbin and minimize energy consumption. Meanwhile, other studies have begun to shift freight modes to low-carbon and efficient transport structures. Lv et al. (2019) found that roadway transport has the biggest increase in carbon emission in the freight transport sector and transforming from roadway traffic to more environmentally friendly transport modes will reduce energy consumption and carbon emission. Through appropriate incentives, Kaack et al. (2018) promoted the transformation of public transport into railway or waterway transport in freight transport, to achieve energy conservation and emission reduction. The above literature focuses on the optimization of transport structure with carbon as the goal but ignores the impact on society and the economy.

The third category is to investigate the optimization of transportation structure from multiple perspectives and multiple objectives. For example, Li et al. (2019b) established a multi-objective model to maximize transportation utility and ecological impact while minimizing generalized cost from the perspective of passenger transport, and then they carried out an optimization analysis. Lin and Wang (2022) construct a multi-objective optimization model for China's transport structure, which can achieve a balance between the constrained environmental, economic, and social objectives and optimize the transport structure using a multi-objective particle swarm optimization algorithm. Zhang et al. (2018) took Shanghai as an example and selected five factors from the perspective of individuals and the public, including travel cost, congestion

degree, occupied area, traffic emission, and operation subsidies, and the objective functions of these factors were converted into satisfaction functions. Meanwhile, a multi-objective programming model was employed to solve the optimal ratio of ground bus and railway transit, and the carbon emission reduction potential was analyzed under different scenarios. In another study, Qian et al. (2021) first quantitatively analyzed the economic benefits, carbon emission intensity, and input–output efficiency of different transport modes in different provinces. Then, to maximize the efficiency of the freight system while reducing carbon emission intensity and not reducing economic output, an optimization model was constructed to obtain the freight structure adjustment scheme of each province.

To sum up, abundant results and relevant experience have been obtained in the research on freight structure, but there are still some limitations.

(1) In most research on transportation structure, there are few studies on the related aspects of multi-objective transportation structure and even fewer studies on the comprehensive consideration of carbon emission and transportation utility value (safety, transport price, speed, and flexibility).
(2) There are few studies on freight structure optimization with the planning of annual carbon peak, but to adjust the current freight structure in China, the planning year should be set to 2030, which is significant to the current freight structure adjustment.
(3) Few studies use intelligent algorithms to solve the multi-objective optimization of freight transportation structure.

Therefore, this paper investigates the current direction of freight structure adjustment in China with carbon emission as the main objective and freight transportation utility value as the auxiliary objective (freight transportation utility value includes safety, transportation price, rapidity, and flexibility). Meanwhile, it is necessary to meet the transportation turnover needs of the whole society, pay attention to the economic and social benefits of transportation, and consider the ecology of the transportation system. The innovations of this paper are as summarized as follows: (1) Consider not only carbon emission reduction but also transport safety, price, speed, and flexibility. (2) In the context of the carbon peak, the freight sharing rate in 2030 is solved to provide theoretical support for freight structure adjustment. (3) Considering the two objectives of carbon emission and transportation efficiency, a fitness function of multi-objective optimization is designed, and the adaptive heuristic layout algorithm is implemented, which improves the convergence speed and accuracy.

Model establishment

Problem description

Since the turnover of other freight is relatively small than that of roadway freight, railway freight, and waterway freight (excluding ocean transportation), this paper only considers roadway freight, railway freight, and waterway freight (excluding ocean transportation) in optimizing the freight structure. Meanwhile, a multi-objective model with transportation carbon emission as the main objective and freight utility value as the secondary objective is constructed to optimize the planning of annual freight structure, and the following two assumptions are made on the model.

Hypothesis 1: Freight transportation is divided into three types: roadway, railway, and waterway (excluding sea transportation).
Hypothesis 2: Different modes of transport have substitutability, and one freight transport mode can be transferred to another through external influencing factors.

Optimization principle

The optimization of freight structure that pays attention to the economic and social benefits of freight should guarantee that the demand for freight turnover of the whole society can be met in the planning year. Meanwhile, the ecological nature of the freight transport system in the planning year should also be considered. The following four principles should be followed in the optimization process.

(1) The optimization results should guarantee that the planned annual freight turnover supply can meet the needs of social development as much as possible;
(2) The coordinated development of different modes of freight transport in the planning year should be considered as much as possible;
(3) The energy consumption of freight transport should meet the requirements of energy saving and emission reduction in the planning year as much as possible;
(4) The optimal freight structure obtained should make the best use of the existing freight infrastructure.

Introduction of constructing multi-objective function

To achieve the carbon peak goal, China needs to adjust the current freight structure to reduce carbon emissions while considering the coordination of the economy, society, and

environment. The multi-objective programming problem is composed of two basic parts: (1) two or more objective functions; (2) several constraints.

Objective function

Goal 1 Minimize freight carbon emissions as much as possible.

By optimizing the freight structure, transportation carbon emission is minimized. This paper mainly studies roadway freight, railway freight, and waterway freight (excluding ocean transportation), so they are calculated separately, as shown in Eq. (1).

$$F_1 = \min \sum_{i=1}^{3} \lambda_i T_i \tag{1}$$

where F_1 represents the sum of carbon emissions generated by each freight mode (unit: million tons); $i(i=1,2,3)$ represents the three freight modes (excluding ocean transportation); T_i represents the turnover of freight mode i (unit: billion tons. km); λ_i represents the carbon dioxide emissions of freight mode i (unit: million tons /billion tons. km).

Goal 2 Maximize the total utility value of freight transportation as much as possible.

The total utility of the freight transport system is influenced by several factors, including the speed of the transport service process, the safety of the means of transport, the reasonableness of the tariff setting, and the level of timeliness. In this paper, four aspects are measured in terms of safety, transportation price, rapidity, and flexibility, and they are calculated separately, as shown in Eq. (2).

$$F_2 = \max \sum_{i=1}^{3} u_i T_i \tag{2}$$

where F_2 represents the freight transport utility value (unit: billion tons. km), and u_i represents the utility value of freight mode i.

Total goal: Minimize carbon emissions and maximize total utility value as much as possible.

$$\max F = -a\left(\lambda_1 T_1 + \lambda_2 T_2 + \lambda_3 T_3\right) + b\left(\mu_1 T_1 + \mu_2 T_2 + \mu_3 T_3\right) \tag{3}$$

In Eq. (3), a and b are respectively the weight coefficients of the freight transportation utility function and traffic carbon emission. They vary according to the preference of the decision subject, and the sum of the decision coefficients equals 1. Since this paper takes the carbon emission of transportation as the main target and the utility value of freight transportation as the auxiliary target, the minus sign is used to transform the minimization problem of transportation carbon emission into a maximization problem. After the above processing, a multi-objective maximization and minimization problem is transformed into a single-objective optimization problem for maximization.

Constraints

(1) Regarding economic development, a low-carbon and efficient freight transport system should meet the needs of socio-economic development, so the total turnover of the three freight modes in the model should satisfy the demand of the whole society in the planning year, as shown in Eq. (4).

$$\sum_{i=1}^{3} T_i \geq D \tag{4}$$

where D represents the upper limit of the total turnover of various modes of freight transportation of the freight demand forecast for the planning year (unit: billion tons. km).

(2) Regarding energy consumption, total freight energy consumption should meet the requirements of the planning year, as shown in Eq. (5).

$$\sum_{i=1}^{3} \rho_i . T_i \leq E \tag{5}$$

where ρ_i represents the unit turnover energy consumption of freight mode i (unit: ten thousand tons of standard coal/ one hundred million tons. km); E represents the upper limit of energy consumption capacity of freight transport in the planned year (unit: ten thousand tons of standard coal).

(3) Regarding carbon dioxide emissions, freight carbon emissions should meet the requirements of the planning year, as shown in Eq. (6).

$$\sum_{i=1}^{3} \lambda_i . T_i \leq E_{co2} \tag{6}$$

where E_{co2} represents the upper limit of carbon dioxide capacity for the planned annual freight transportation (unit: ten thousand tons).

(4) As for the use of transport networks, high-quality transport systems should rationally use existing transport infrastructure. There is a fixed proportional relationship between the proportion of different modes of freight transportation in the total

traffic volume and the proportion of line length in the total length of the transportation system. This ratio is called the utilization rate of this freight transportation mode. A higher traffic network utilization degree than the reasonable utilization rate can ensure the effective utilization of existing traffic infrastructure, as shown in Eq. (7).

$$\frac{\frac{T_i}{L_i} / \sum_{i=1}^{3} \frac{T_i}{L_i}}{S_i / \sum_{i=1}^{3} S_i} \geq \alpha_i \tag{7}$$

where L_1 represents the average transportation distance of freight mode i (unit: km); S_i represents the line length of freight mode i (unit: km); α_i represents the line utilization of freight mode i.

Algorithm

Introduction of the traditional GA

Sample data generation and processing

The independent variable T_i of the optimization model constructed in this paper is the freight turnover of railways, roadways, and waterways, which is produced by generating a random number between the upper and lower bounds. Meanwhile, the value and dimension of each freight turnover are inconsistent, so there will be certain errors. To eliminate the influence of this error on the experimental results, the variables need to be normalized. After the original data is standardized, each index has the same order of magnitude, which is conducive to comprehensive comparative evaluation. Common normalization methods include maximum–minimum normalization (He et al. 2021) and zero-mean (Park and Pyo 2022) normalization. The maximum–minimum normalization is to standardize the maximum and minimum values in the data column, and the normalized values are between [0, 1]. The calculation method is to calculate the difference between the function and the minimum value of the column and then divide the maximum value of the column minus the minimum value of the column, as shown in Eq. (8).

$$T_{i,j}^{*} = \frac{T_{i,j} - T_{i,\min}}{T_{i,\max} - T_{i,\min}} \tag{8}$$

where $T_{i,j}$ is the value of sample j of freight mode i; $T_{i,\min}$ is the minimum value of freight mode i; and $T_{i,\max}$ is the maximum value of freight mode i.

Chromosome coding

After normalization, the variables of the optimization model are transformed into real numbers. In this paper, floating point coding (Szabo and Dobroka 2018) is adopted, and a chromosome is used to represent the candidate solution to the problem, thereby forming the basic work of the genetic algorithm.

Fitness calculation

After dimensionality reduction, the objective function of this paper is transformed into the fitness function, which is proportional to the objective function. The higher the fitness value of the individual, the higher the objective function value, and the better the solution. For individuals that meet the constraints, the fitness value is calculated using Eq. (9). For individuals that do not meet the constraints, the fitness function value is set to 0 to guarantee that these individuals are not inherited.

$$\max F = -a\beta_1\left(\lambda_1 T_1 + \lambda_2 T_2 + \lambda_3 T_3\right) + b\beta_2\left(\mu_1 T_1 + \mu_2 T_2 + \mu_3 T_3\right) \tag{9}$$

where F is the fitness function; T_i represents the turnover of freight mode i (unit: hundred million tons. km), β_1 represents the dimensionless coefficient of objective function 1, and β_2 represents the dimensionless coefficient of objective function 2.

Selection operator

The process of selecting excellent individuals and eliminating inferior individuals from a group is called selection. The purpose of selection is to directly inherit the optimized individual (or solution) to the next generation or to generate new individuals through pairing crossover and then inherit them to the next generation. Several commonly used selection methods include roulette selection (Zhang and Han 2022), random competition selection (Xiao 2022), and non-replay random selection. This paper uses roulette selection in the selection process.

Operations of crossover and mutation

The crossover operation of the adaptive genetic algorithm refers to the exchange of some genes between two paired chromosomes, thereby forming two new individuals. The commonly used crossover operations include single-point crossover (Abdel-Basset et al. 2022), two-point crossover (Farmand et al. 2021), and multi-point crossover (Manzoni et al. 2020). Since the variable dimension of the optimization model constructed in this paper is not large, the single-point crossover can effectively change individual traits. By

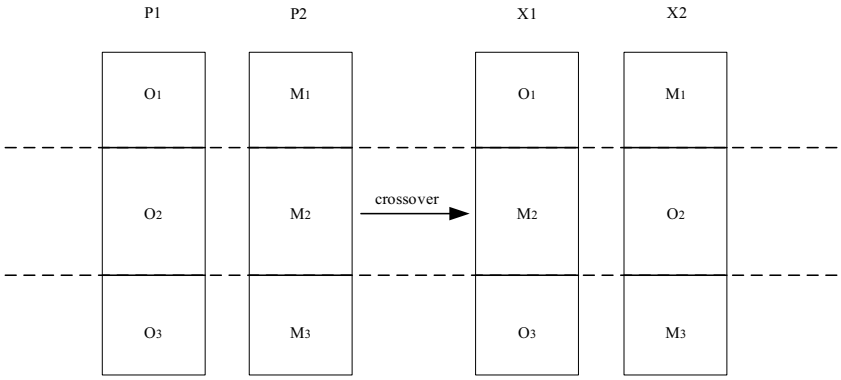


Fig. 3 The schematic diagram of the crossing process

generating random integers, the gene crossover position is determined to obtain the offspring individual, as illustrated in Fig. 3.

In the genetic algorithm, the mutation operation refers to replacing some genes at certain loci in the coding string of an individual chromosome with other alleles to form a new individual. In this paper, the method of random number generation is adopted to determine the gene fragments of variation. Meanwhile, in the early stage of evolution, uniform variation is adopted to increase the range of variation by generating random numbers to accelerate the search for the optimal solution. While in the late stage of evolution, non-uniform variation is adopted to form small perturbations around genes, which is more conducive to the convergence of the adaptive genetic algorithm. Non-uniform variation indicates that when the genetic algebra is small, the square term is close to 1, and thus the variation rate remains almost unchanged. When the genetic algebra is large, the closer the square term is to 0, the lower the variation rate. The advantage of using the non-uniform mutation operation is that while the mutation rate is high at the beginning, the mutation rate will gradually decrease as the number of inherited algebras increases, thus guaranteeing that better genes are passed on. The calculation formula of non-uniform variation is shown in Eq. (10).

$$x'_{i,j} = \begin{cases} x_{i,j} + \left(x_{i,j} - x_{\max}\right) \bullet r_1 \bullet \left(1 - \frac{g}{G_{\max}}\right)^2, \mathrm{r}_1 > 0.5 \\ x_{i,j} + \left(x_{\min} - x_{i,j}\right) \bullet r_2 \bullet \left(1 - \frac{g}{G_{\max}}\right)^2, \mathrm{r}_1 \le 0.5 \end{cases} \tag{10}$$

where $x'_{i,j}$ is the sample value after variation; $x_{\max}$ and $x_{\min}$ are the upper and lower bounds of predefined samples, respectively; r_1 and r_2 are random numbers between 0 and 1; g and $G_{\max}$ are the current genetic algebra and the largest genetic algebra, respectively. $(x_{i,j} - x_{\max})$ and $(x_{\min} - x_{i,j})$ determine the direction of variation; $\left(1 - \frac{g}{G_{\max}}\right)^2$ determines the intensity of variation.

In the optimization model constructed in this paper, the number of populations in each generation is set to 50, and the maximum number of evolutions is 200. When the genetic evolution reaches the maximum number, the evolution stops. The flow chart of the simulation process is shown in Fig. 4.

Introduction of AGA

The AGA has the same steps as the GA. This paper makes improvements to the traditional GA by using adaptive crossover rate and mutation rate. The calculation process of adaptive crossover rate and mutation rate is shown in Eqs. (11) and (12).

$$pc = \begin{cases} k_1 \frac{f_{\max} - f'}{f_{\max} - f_{avg}}, f' \ge f_{avg} \\ k_2, \quad f' < f_{avg} \end{cases} \tag{11}$$

where $f_{\max}$ is the highest fitness of the population; f_{avg} is the average fitness of the population; f is the higher fitness between the parents involved in the crossover; k_1 and k_2 are constants, and they are set to 1 in this paper. When f' is lower than the average value, the crossover rate is greater than 1, and cross-exchange must occur. On the contrary, when f' is higher than the average value, the crossover rate is less than 1, and reducing the crossover rate is conducive to maintaining good traits.

$$pm = \begin{cases} k_3 \frac{f_{\max} - f}{f_{\max} - f_{avg}}, f' \ge f_{avg} \\ k_4, \quad f' < f_{avg} \end{cases} \tag{12}$$

where f is the individual fitness that determines whether to mutate; k_3 and k_4 are constants, and they are set to 1 in this paper. When f' is smaller than the average value, the variation rate is greater than 1, and the variation is inevitable. On the contrary, when f' is larger than the average value, the variation rate is smaller than 1, which reduces the variation rate and facilitates the maintenance of excellent traits.

Determination of model parameters in 2030

During the general debate of the 75th session of the United Nations General Assembly, China announced that it will adopt stronger policy measures to peak carbon dioxide

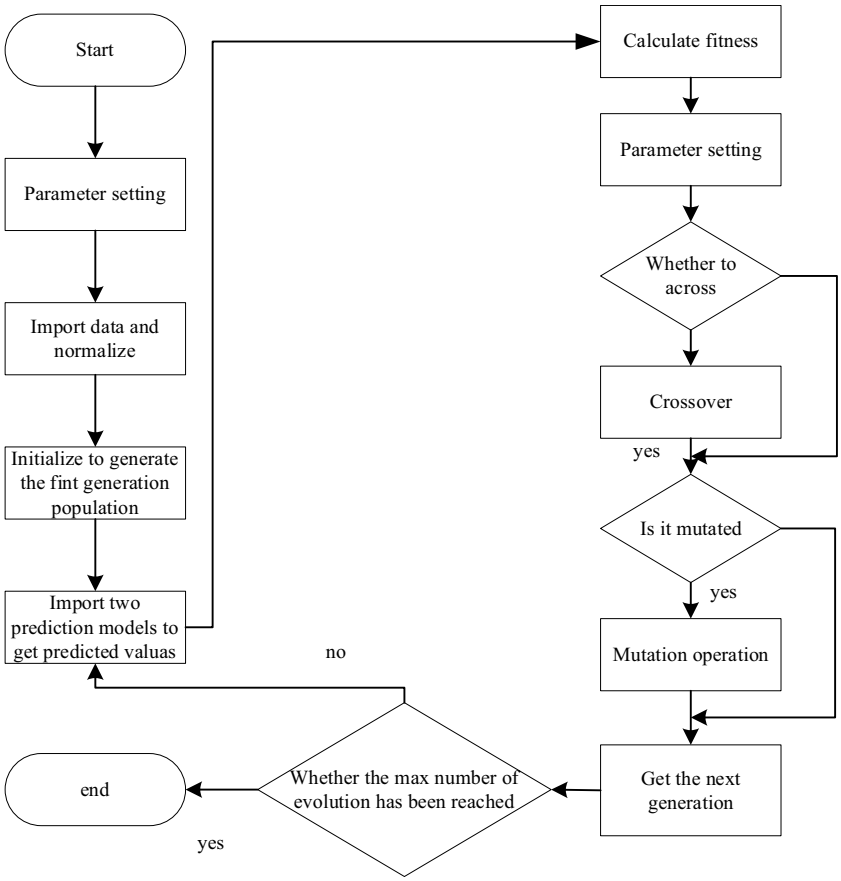


Fig. 4 The flowchart of the traditional GA

emissions by 2030 and achieve carbon neutrality by 2060. It can be seen that the freight structure in 2030 has important guiding significance for current freight adjustment in China. Therefore, this paper sets the planning year to 2030 to obtain the ideal freight structure in 2030, thereby guiding current freight structure adjustment in China.

Determination of utility value

Considering the characteristics of the freight industry, the utility value of various freight modes is considered from four aspects: safety, freight, speed, and flexibility. The highest score of each indicator is 1 point. Due to the lack of scientific scoring based on the author's subjective evaluation, the research values of other scholars are cited, as shown in Table 1.

According to the score in Table 1, this paper calculates the utility value of different freight modes. The freight rate, flexibility, and rapidity of different freight modes are considered a part respectively and calculated by addition, while the safety and the three parts are independent of each other and calculated by multiplication, as shown in Eq. (13).

$$u = \left(w_1 + w_2 + w_3\right) \times w_4 \tag{13}$$

where w_1 represents the freight rate, w_2 represents flexibility, w_3 represents rapidity, and w_4 represents safety. The utility values of roadway freight, railway freight, and waterway freight are represented by u_1, u_2, and u_3 respectively. Through calculation, we have $u_1 = 2.5$, $u_2 = 2.1$, and $u_3 = 2.0$.

Table 1 The utility value of different freight modes

Indicator	Safety	Freight rate	Quickness	Flexibility	Source
Roadway	1	0.5	1	1	(Li 2012)
Railway	1	0.7	0.7	0.7	(Li 2012)
Waterway (excluding ocean transportation)	1	1	0.5	0.5	(Li 2012)

Selection of weight coefficient

Carbon emission reduction is a challenging issue that needs to be solved urgently. With the goal of carbon peak, carbon emission reduction becomes crucial. This paper constructs a dual-objective model based on traffic carbon emission and freight utility value. When selecting the weight coefficient, this paper believes that in the context of carbon peak, carbon emission should be considered more, but the weight coefficients of the two objectives cannot be too different. Therefore, the weight coefficients of freight utility value and traffic carbon emission are set to 0.4 and 0.6, respectively.

Determination of energy consumption per unit turnover

In this paper, the energy consumption per unit turnover of roadway freight, railway freight and waterway freight (excluding ocean transportation) is obtained by consulting relevant literature. As shown in Table 2.

Determination of carbon dioxide carbon emission factor

In this paper, carbon dioxide emissions per unit turnover of roadway freight, railway freight and waterway freight (excluding ocean transport) are obtained by consulting relevant literature, as shown in Table 3.

The upper limit reached in 2030

According to the prediction given by experts of the Chinese Academy of Sciences, the carbon emission from transportation in 2030 is about 100,000 tons, and the energy consumption of freight transportation is about 48 million tons of standard coal. According to Chinese statistical years with the average growth coefficient method, the total turnover of freight transport in 2030 is 203563 billion tons. km, as shown in Table 4.

Reasonable utilization rate of freight transportation lines

A reasonable utilization rate of freight lines makes the utilization rate in 2030 higher than that at present, thereby guaranteeing the effective use of the existing transportation infrastructure in 2030. China was affected by the COVID-19 pandemic from 2020 to 2022, which reduced the utilization rate of freight transportation infrastructure. To formulate a higher standard for the optimization of freight structure in 2030, the average reasonable utilization line rate from 2013 to 2019 was selected to make the utilization rate of freight transportation infrastructure in 2030 greater than that before

Table 2 The energy consumption per unit turnover of the transportation structure in 2030

Transportation structure	Energy consumption in 2030	Unit	Source
Roadway	0.3260	10000 tons of standard coal/100 million tons. km	(Wang 2021)
Railway	0.0486	10000 tons of standard coal/100 million tons. km	(Wang 2021)
Waterway (excluding ocean transportation)	0.0780	10000 tons of standard coal/100 million tons. km	(Wang 2021)

Table 3 The carbon dioxide emission factor per unit turnover of the transportation structure in 2030

Transportation structure	Total carbon emission in 2030	Unit	Source
Roadway	0.705	10000 tons of standard coal/100 million tons. km	(Wang 2021)
Railway	0.119	10000 tons of standard coal/100 million tons. km	(Wang 2021)
Waterway (excluding ocean transportation)	0.170	10000 tons of standard coal/100 million tons. km	(Wang 2021)

Table 4 The caps for each indicator in 2030

Category	Quantities to be reached in 2030	Unit	Source
Transportation carbon emission	1000000000	Tons	(Mai 2021)
Energy consumption for transportation of freight	480	Million tons of standard coal	(Mai 2021)
Total freight turnover	203563	Billion tons km	https://item.kongfz.com/book/35802361.html

Table 5 The line utilization rates of roadway freight, railway freight, and waterway freight (excluding ocean transportation) from 2013 to 2019

Transportation structure	2013	2014	2015	2016	2017	2018	2019	Mean value	Source
Roadway	0.82	0.83	0.83	0.83	0.84	0.84	0.80	0.82	China statistical yearbook
Railway	4.45	3.78	3.33	3.14	3.14	3.13	3.65	3.52	China statistical yearbook
Waterway (excluding ocean transportation)	4.49	4.60	5.09	5.14	5.04	5.04	6.07	5.07	China statistical yearbook

Table 6 Freight turnover and freight structure in 2030

Freight transport mode	Freight turnover (unit: billion tons. km)	Freight structure
Roadway	88766.63	42.42%
Railway	49499.18	23.65%
Waterway (excluding ocean transportation)	70991.02	33.93%

the epidemic. Let the proportion of freight transported in a certain transport mode in the total traffic volume be w_i and the proportion of line length in the total length be z_i; then, w_i/z_i represents the reasonable utilization rate of this transport mode. Based on the data released by China Statistical Yearbook, reasonable utilization rate and average reasonable utilization rate of different transportation modes in China from 2013 to 2019 were obtained by calculation, and the results are shown in Table 5.

Results and discussions

Results

With the help of MATLAB(R2017a) and based on AGA, the ideal freight turnover of roadway freight, railway freight, and waterway freight (excluding ocean transportation) in 2030 is 88,766.63, 49,499.18, and 70,991.02 (unit: billion tons. km), respectively. Accordingly, the freight structure of roadway, railway, and waterway (excluding sea transportation) is 42.42%, 23.65%, and 33.93%, respectively, and the results are shown in Table 6.

Discussions

Comparative analysis of freight structure before and after optimization

In this paper, the average value of freight structure from 2016 to 2021 is selected to represent the current freight structure of China, i.e., the freight structure before optimization, and the optimized freight structure is the freight structure in 2030.

As China was hit by COVID-19 at the end of 2019, the freight structure was severely affected, especially waterway freight (excluding ocean transportation) (Lu et al. 2023). The freight sharing rate of waterways (excluding ocean transportation) directly dropped from 35.72% in 2019 to 14.77% in 2021, showing a decrease of about 50%, which is mainly related to the epidemic. Now, with the relaxation of the epidemic policy, all industries in China have gradually recovered, and the freight turnover of waterways (excluding ocean transportation) will experience a substantial increase, so the freight structure during the epidemic cannot reflect the current situation. Therefore, in this paper, the average value of the freight structure from 2016 to 2021 is chosen to represent the freight structure before optimization, as shown

Table 8 Comparison of the freight structure before and after optimization

Freight transport mode	Before optimization	After optimization	Change
Roadway	50.49%	42.42%	−8.07%
Railway	22.72%	23.65%	+0.93%
Waterway (excluding ocean transportation)	26.80%	33.93%	+7.13%

Table 7 The freight share of roadway, railway, and waterway (excluding ocean transportation) from 2016 to 2021

Freight transport mode	2016	2017	2018	2019	2020	2021	Mean value
Roadway	49.20%	48.65%	48.40%	42.68%	56.43%	57.54%	50.49%
Railway	19.17%	19.64%	19.58%	21.60%	28.62%	27.68%	22.72%
Waterway (excluding ocean transportation)	31.63%	31.71%	32.02%	35.72%	14.95%	14.77%	26.80%

in Table 7. The comparison of freight structure before and after optimization is presented in Table 8.

It can be seen from Table 8 that compared with the current freight structure of China, the roadway freight sharing rate in 2030 will decrease by 8.07%, and the railway freight sharing rate and the waterway (excluding ocean transportation) freight sharing rate will increase by 0.93% and by 7.13% respectively. Overall, the reduction of the roadway freight turnover sharing rate is consistent with the requirements of China's roadway freight, which is conducive to achieving the goal of carbon peak. Although the freight sharing rate of railway freight turnover increases, the growth rate is not obvious, and this is mainly because the large increase of waterway (excluding ocean transportation) freight offsets the railway freight sharing rate. The reason why waterway (excluding ocean transportation) freight turnover has increased significantly is mainly that the current waterway (excluding ocean transportation) freight turnover in China is much lower than that before the epidemic. The freight structure in 2030 obtained in this paper is similar to that predicted by the Ministry of Transport of China.

Comparative analysis of energy consumption and carbon emission before and after optimization

With the development of the industrial economy, the dramatic increase of population, and the unlimited rise of human desires, more and more problems have emerged in resources and climate, and energy consumption and carbon dioxide emissions are increasing (Khan et al. 2022, Zou and Zhang 2020). Currently, one of the important criteria to measure the freight transport structure in China is whether the freight structure is conducive to energy saving and emission reduction. Then, energy consumption and carbon emission before and after freight structure optimization are compared and analyzed in this paper.

Compared with the current total freight turnover in China, the total freight turnover in 2030 will increase a lot, so it is not reasonable to directly compare the carbon emission between the two time points. The idea of energy consumption and carbon emission before the optimization is to select the total freight turnover in 2030 to guarantee that the total freight turnover before and after optimization is constant. The current freight structure in China can be divided into freight transportation modes of roadway, railway, and waterway (excluding ocean transportation), as shown in Fig. 5. Then, multiplied by carbon emission per unit turnover, the energy consumption and carbon emission before optimization are calculated. Finally, the optimized energy consumption and carbon emission are calculated by multiplying the freight turnover in 2030 by the respective energy consumption intensity and freight carbon emission factors (Tables 2 and 3). Figures 6 and 7 show the energy consumption and carbon emission before and after optimization. The total energy consumption and total carbon emission before and after optimization are presented in Table 9.

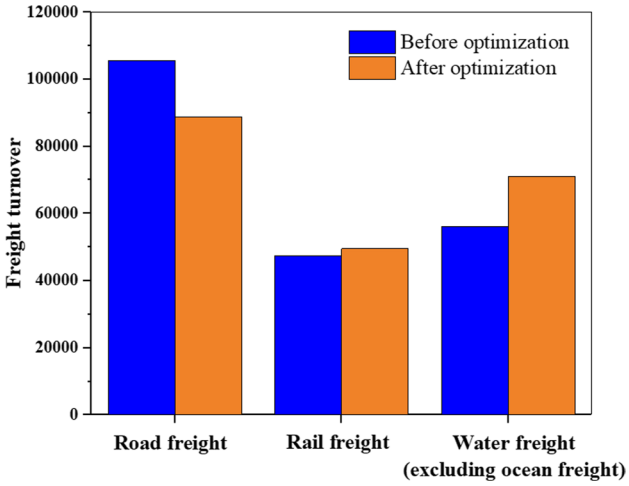


Fig. 5 The freight turnover before and after optimization (unit: billion tons. km)

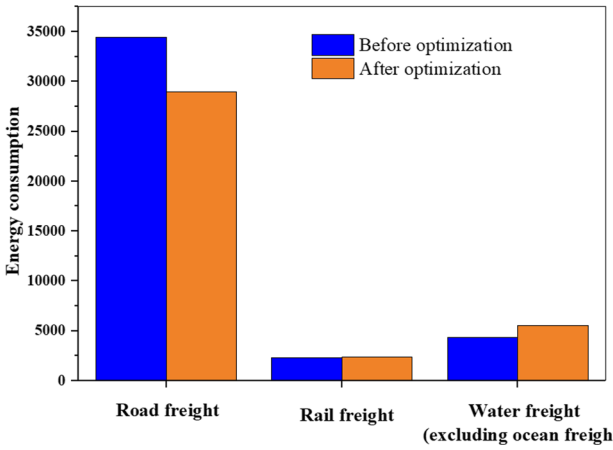


Fig. 6 The energy consumption before and after optimization (unit: tons)

It can be seen from Table 9 that the energy consumption and carbon emission before optimization are 441,280,300 tons and 896,772,900 tons of standard coal, respectively. After optimization, the energy consumption and carbon emission in 2030 will be 368,888,800 tons and 805,393,500 tons of standard coal, respectively. After optimization, the energy consumption and carbon emission are reduced by 42,471,500 tons and 91,379,400 tons of standard coal, by 10.3% and 10.2%, respectively. For the freight industry, reasonable adjustment of the freight structure will help to save energy and reduce emissions, which is conducive to achieving the carbon peak goal of the country. The results of this study are similar to those of Guo et al. (2014), who

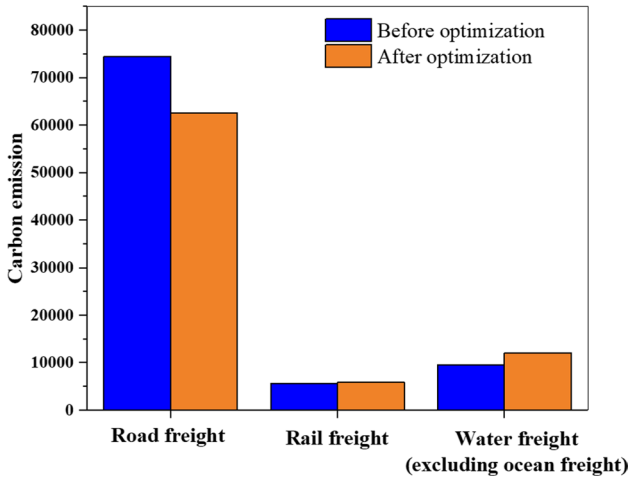


Fig. 7 The carbon emission before and after optimization (unit: tons of standard coal)

illustrated that transportation structure optimization helps to reduce carbon emissions.

Comparative analysis of algorithm performance

After several operations, this paper finally obtains a comparison chart that can represent the average level of the two algorithms. Then, the AGA and the traditional GA are compared and analyzed in terms of convergence time and accuracy.

The average fitness curves of the populations of the conventional GA and AGA in Fig. 8 indicate that the fitness value of AGA decreases rapidly from 0 to 10 generations. From 10 to 50 generations, although the fitness value fluctuates, it shows an overall convergent trend, and the maximum fitness value reaches 0.4168. Under the same parameters, the traditional GA is adopted, and the fitness value becomes larger from 0 to 25 generations. However, from 25 to 38 generations, the fitness value decreases rapidly, and then it reaches the extreme point repeatedly from 38 to 100 generations until the convergence is reached around 100 generations. The maximum fitness value is 0.4032.

In this paper, a maximization and minimization multi-objective problem is transformed into a single-objective maximization problem. The fitness function is positively correlated with the value of the objective function, so the larger the value of the fitness function is, the better the solution is obtained. The maximum average fitness value of AGA is larger than that of the traditional GA. Therefore, AGA has higher accuracy than the traditional GA. In terms of convergence speed, AGA achieves convergence in 50 generations, while the traditional GA achieves convergence in 100 generations. By contrast, AGA has better precision than the traditional GA. To sum up, AGA outperforms the traditional GA in terms of convergence speed and accuracy. The results of this study are similar to those of Ding et al. (2022).

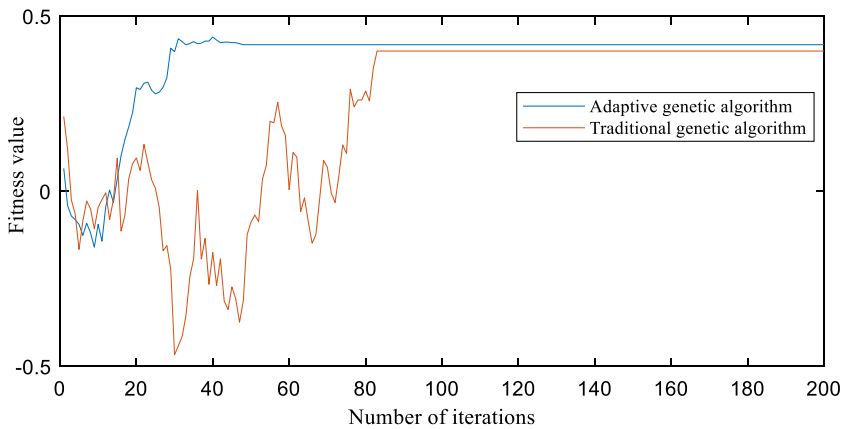
Sensitivity analysis

Sensitivity analysis is a method to study and analyze the sensitivity of a system or model state to system parameters. It is usually used to analyze which parameters have a great

Table 9 Comparison of the total energy consumption and total carbon emission before and after optimization

Category	Before optimization	After optimization	Total reduction	Percentage decrease
Energy consumption (unit: tons)	896,772,900	805,393,500	91,379,400	10.2%
Carbon emission (unit: tons of standard coal)	441,280,300	368,888,800	42,471,500	10.3%

Fig. 8 The comparison of average fitness curves of the populations



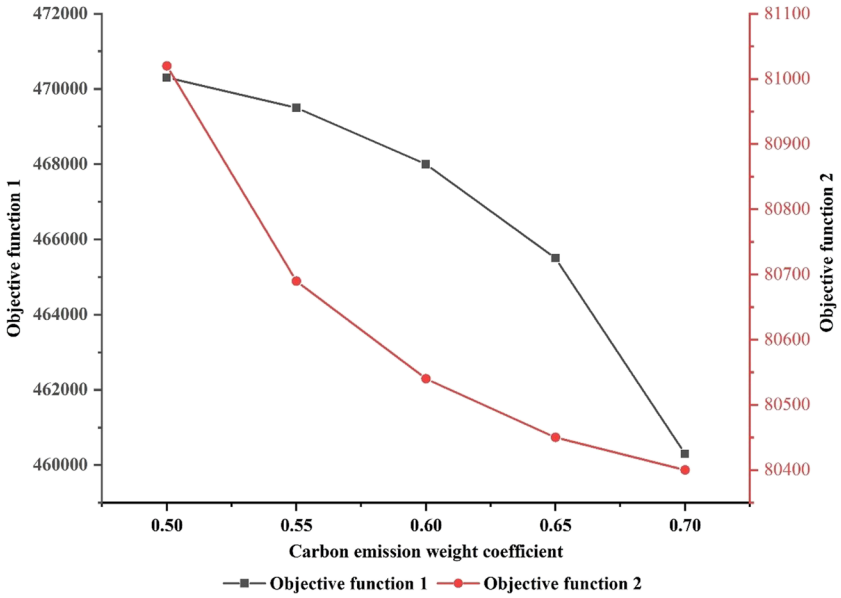


Fig. 9 The change curve of objective function 1and objective function 2 as the carbon emission weight coefficient increases

influence on the system or model, and it is currently used in many fields such as machinery (Li et al. 2019c), energy (Sivapirakasam et al. 2020), and medicine (Svensberg et al. 2019). To analyze the influence of different weight coefficients on objective function 1 and objective function 2, five groups of weight coefficients are set in this paper, and they range from 0.5 to 0.7 with a step length of 0.05. The experimental results are presented in Fig. 9.

This paper constructs a model with transportation carbon emission as the main objective and freight utility value as the secondary objective. The weight coefficients of the freight utility value and carbon emission value are set to 0.4 and 0.6 respectively. As the carbon emission weight coefficient increases, both the freight utility value and the carbon emission decrease.

Let the weight coefficients be (0.45, 0.55), (0.40, 0.60), and (0.35, 0.65) for scenario 1, scenario 2, and scenario 3, respectively. By comparing scenario 1 and scenario 2, it can be seen that scenario 1 is more sensitive to carbon emission than scenario 2, and scenario 2 is more sensitive to freight utility value than scenario 1. By comparing scenario 2 and scenario 3, it can be seen that scenario 2 is more sensitive to carbon emission than scenario 3, and scenario 3 is more sensitive to freight utility value than scenario 2. Besides, it can be seen from the slope of Fig. 9, the utility value of freight transportation has an increasingly higher slope, which represents the increasing sensitivity; meanwhile, the carbon emission has an increasingly lower slope, which represents the decreasing sensitivity.

In summary, as the weight coefficient of carbon emission increases, the utility value of freight transportation consistently decreases, and the sensitivity increases. As the carbon emission weight coefficient increases, carbon emission keeps decreasing, and the sensitivity decreases.

Conclusions and policy suggestions

This paper constructs a model that takes transportation carbon emission as the main target and freight transportation utility value as the auxiliary target. The constructed model satisfies the constraints of freight turnover in the whole society, freight economic and social benefits, and the ecological constraints of the freight system. With MATLAB, the freight turnover of roadways, railways, and waterways in 2030 is solved by using the adaptive genetic algorithm. The results show that the model solves the freight structure in 2030 well, which makes it easier for decision makers to adjust China's current freight structure in the process of reaching the carbon peak in 2030. At the same time, this method solves the model well and is better than the traditional genetic algorithm. Therefore, in order to better achieve the goal of carbon peak, our government should introduce more policies to support the adjustment of freight structure; in terms of road freight transportation, law enforcement should be strengthened to promote the development of new energy vehicles; complementary advantages pay attention to the development of railway freight and waterway freight.

Author contribution Hang Ke almost was responsible for the whole of the paper; Guangying Xu guided the thesis writing; Chuntang Li, Jing Gao, Xinrui Xiao, Xin Wu, and Quanwei Yan were responsible for modifying the format and searching data.

Data availability Not applicable.

Declarations

Ethics approval and consent to participate Not applicable.

Consent for publication Not applicable.

Competing interests The authors declare no competing interests.

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